

A Backreacting Phase Clock with Fixed Momentum in Pure-Weyl Gravity

Fixed-Coupling Rigidity and Causal BV Cartan Analysis of the Helical Stabilizer

GPT-5.6.sol*

Asger Alstrup Palm†

Theorem-frozen version, July 2026

Abstract

We study a backreacting phase clock whose internal momentum is nonzero but first-order rigid within each fixed theory. On $\mathbb{R} \times S^3$ with a Berger spatial metric we exhibit a one-parameter incidence family across the scalar coupling, consisting of exact, smooth, non-conformally-flat solutions sourced by two standard-sign conformal scalars. Their rotating phase is an everywhere timelike clock, the quartic potential is positive, and the internal clock momentum is nonzero. Nevertheless, at fixed couplings the linearized lapse constraint and compact spatial averaging force its variation to vanish on every smooth solution tangent. The covariant identity $\Omega_{\text{total}}(\delta, \mathcal{L}_D) = \omega \delta Q_R$ therefore makes the cylinder generator D a presymplectic degeneracy on this declared linearized phase space. This proves infinitesimal nullity, not integrability of a nonlinear quotient.

At one exact rational member we analyze the associated stationary helical stabilizer homologically. The complete gauge-fixed classical BV presentation has 54 rows and a support-local cyclic retract onto a 26-row retained complex. Advanced and retarded chain contractions lift to all rows. Differentiating the covariant master action gives complete arbitrary-input four-dimensional operations q_2 and q_3 , with the required L_∞ identities, cyclicity and local K -equivariance. The unary and binary Cartan primitives extend to a cyclic ternary primitive: the arity-three source closes by the graded Jacobi identity, and a causal raw primitive becomes cyclic under the exact C_4 Reynolds projector. Thus the $K = D - \omega R$ action is causally null-homotopic through arity three. The co-rotating conjugation is essential: the exported unary operator e_0 is the action of K , whereas the original cylinder translation D is affine about the rotating background and has a nonzero zeroth Taylor component. No affine D -Cartan theorem is claimed. An independent reduced-action calculation crosses the producer boundary in eight lapse and radial/Weyl fourth derivatives, agreeing with sixteen ordered coefficients of the frozen q_3 payload; it is not a second full derivation.

The result is classical: its momentum-rigidity and linear-nullity statement holds pointwise along the displayed incidence branch after fixing the couplings, while its all-row BV Taylor statement is proved at one exact rational representative. It is not an all-orders nonlinear theorem, a Hadamard construction, a quantum-master-equation result, an anomaly calculation, or a claim that a nonzero clock momentum is a freely variable boundary charge.

*OpenAI model and principal programme author.

†Programme orchestrator and corresponding human contact; Honorary Professor, DTU Compute, Technical University of Denmark. asger@area9.dk.

Authorship and computational inputs

The named model author produced the mathematical programme, derivations, verification code, referee responses, and manuscript. Asger Alstrup Palm commissioned and directed the work, initiated the questions, and served as non-technical orchestrator and corresponding human contact; he claims no technical contribution. Repository scripts generate the certificates and sparse coefficient tables used as computational inputs. They are not additional authors. This article has no generated $\text{\TeX}$ input. This attribution describes the research archive. Any submission to a venue with a different authorship policy requires a separate accountability review; that review is not represented by the present working draft.

Contents

1	Question, phase space and theorem boundary	2
2	A positive clock on an exact Berger background	4
3	Nonzero clock momentum and vanishing tangent charge	6
3.1	The fixed-coupling slice and the lapse normalization	6
4	Gauge explanation and relational evolution	8
5	The complete causal BV complex	8
6	Action-derived interaction tensors	10
7	Causal K-Cartan contraction through arity three	11
7.1	Support statement	13
8	Interpretation, failures and open directions	13
9	Reproducibility and claim map	18
9.1	Charged physical time and conditional event maps	19

1 Question, phase space and theorem boundary

There are two superficially conflicting requirements for a relational clock. Its phase must evolve and possess a nonzero conjugate momentum, yet the total Hamiltonian generator should remain a constraint on a closed universe. A matter phase with zero momentum is not a clock, while a freely variable nonzero Hamiltonian charge would make time translation a physical symmetry rather than proper gauge. The distinction is made by the pullback of the charge variation to the allowed solution space, not by the background value of the charge. We use the BV description of gauge systems [2, 3], the covariant phase-space pairing [7], and causal chain contractions in the sense of Green-hyperbolic complexes [17].

We work on

$$M = \mathbb{R} \times S^3 \tag{1}$$

with no spatial boundary. Couplings are fixed. Write $\mathcal{E}_{\alpha_B, \lambda}(\Phi) = 0$ for the covariant equations and $\bar{\Phi}$ for the Berger solution. The charge theorem is evaluated on the pre-reduced smooth tangent

space

$$\mathcal{Z}_{\bar{\Phi}}^{\alpha_B, \lambda} := \ker D\mathcal{E}_{\alpha_B, \lambda}|_{\bar{\Phi}} \subset \Gamma^\infty(M, F), \quad \delta\alpha_B = \delta\lambda = 0, \quad (2)$$

including the linearized constraints. We test whether the D vector lies in the radical of the covariant presymplectic form on this space; we do not quotient by D in the definition. The nonlinear statement is instead a formal Taylor identity for the classical BV vector field through arity three at the same background. These are related statements, but they are not identical.

Let $D := \partial_t$ denote cylinder-time diffeomorphism and let R generate the global, ungauged internal $O(2)$ rotation of the scalar doublet. On the background,

$$\mathcal{L}_D \bar{g} = 0, \quad \mathcal{L}_D \bar{T} = \omega R \bar{T}, \quad K := D - \omega R, \quad \mathcal{L}_K \bar{\Phi} = 0. \quad (3)$$

Thus the solution is stationary in stress energy but not pointwise fixed by D on the scalar fields. The orbit of D is helical in spacetime times the internal target. The geometric theorem below tests the diffeomorphism generator D . The based Taylor BV theorem instead concerns the stabilizer K , the generator that actually fixes the expansion point; this is the standard twisting distinction for a symmetry at a non-fixed background [15]. The global rotation R remains ungauged.

Proposition 1.1 (Generator conjugation). *Let $U(t) = \exp(\omega t R)$ and write the scalar in co-rotating variables as $T = U(t)((\rho, 0) + \psi)$. Then*

$$U^{-1} \mathcal{L}_D T = \partial_t \psi + \omega R \psi + \omega R(\rho, 0), \quad (4)$$

$$U^{-1} \mathcal{L}_K T = \partial_t \psi. \quad (5)$$

Thus raw D has a nonzero zeroth Taylor component $L_D^{(0)} = \omega R(\rho, 0)$ and unary component $L_D^{(1)} = e_0 + \omega R$ in dressed coordinates. By contrast, $L_K^{(0)} = 0$ and $L_K^{(1)} = e_0$. The frozen all-row operation previously labelled “helical D ” is therefore the K action.

Proof. The convention $R(T_1, T_2) = (-T_2, T_1)$ gives $\partial_t U = \omega R U$. Substitution into $D(U((\rho, 0) + \psi))$ gives (4); subtracting ωR gives (5). The coefficientwise 54-row rule is $e_0 I_{54}$, so it realizes the second identity. The strict certificate is the generator-conjugation audit in the computational supplement. $\square$

Theorem 1.2 (Fixed-coupling rigidity of the Berger clock momentum). *Fix $\alpha_B > 0$. Every point of the positive Berger incidence branch in Section 2 carries a smooth standard-sign conformal-scalar phase clock with positive quartic potential, timelike phase gradient and nonzero internal momentum $Q_R > 0$. After fixing (α_B, λ) , however,*

$$dQ_R|_{\mathcal{Z}_{\bar{\Phi}}^{\alpha_B, \lambda}} = 0. \quad (6)$$

Proof. The exact background equations and pointwise matter inequalities are derived in Section 2. The nonzero charge is (23); the fixed-coupling lapse identity and the compact-averaging extension to every smooth tangent are proved in Proposition 3.1. $\square$

Corollary 1.3 (Linear presymplectic nullity of D). *On the declared smooth fixed-coupling linearized solution tangent,*

$$\Omega_{\text{total}}(\delta, \mathcal{L}_D) = 0 \quad (7)$$

for every allowed tangent δ .

Proof. Combine Theorem 1.2 with the covariant identity (25); the closed S^3 leaves no boundary contribution. $\square$

Theorem 1.4 (Causal BV Cartan theorem for the helical stabilizer). *At the exact rational representative (20), the complete 54-row gauge-fixed classical BV complex admits advanced and retarded chain contractions with causal support. If q_1, q_2, q_3 are its first three action-derived Taylor operations, there are cyclic causal components $\iota_K^{(1)}, \iota_K^{(2)}, \iota_K^{(3)}$ such that*

$$([Q, \iota_K] - L_K)^{(n)} = 0, \quad n = 1, 2, 3. \quad (8)$$

The theorem neither constructs the affine zeroth-arity D-Cartan equation nor asserts convergence or an integrated nonlinear quotient.

Proof roadmap. Section 5 constructs the typed Green contractions and their support-local all-row lift. Section 6 derives and audits q_2, q_3 and K -equivariance. Section 7 proves the Hom-space recursion, ternary closure, cyclicity and causal-hull support. $\square$

Theorem 1.2 and Corollary 1.3 are certified by P09-C1–P09-C4. Theorem 1.4 is certified by P09-C5–P09-C10 together with the independent generator-conjugation audit. Neither theorem extends to a renormalized quantum theory. The manuscript status is `THEOREM.FROZEN`. Its ten-claim ledger contains only the gravity-clock background, charge, causal BV and K -Cartan results stated above. Maxwell signal and observer-apparatus results are deliberately excluded from both main theorems.

Proof architecture. The geometric layer derives the exact background and charge identity from the action. The analytic layer constructs the retained Green homotopies and lifts them through a support-local deformation retract. The algebraic layer differentiates the master action and solves the first three Cartan recurrences. Computer algebra certifies the instantiated curved coefficients and all-row signs; it does not replace the invariant definitions or the compact-averaging argument.

Novelty and relation to established machinery. Relational and complete observables, covariant phase-space charges, linearization stability, BV–BFV theory, cyclic homotopy algebras, homotopy transfer, Cartan homotopies and Green homotopies are established frameworks [4, 5, 6, 8, 10, 11, 12, 13, 14, 17]. The contribution claimed here is their explicit conjunction in one action-derived example: a positive fully backreacting Berger clock, a fixed-coupling momentum-rigidity theorem, a complete gauge-fixed classical BV presentation, and a cyclic causally supported Cartan primitive for its helical stabilizer through q_3 . To our knowledge the specific Berger/Weyl/scalar construction is new; no priority claim is made for any of the general frameworks.

2 A positive clock on an exact Berger background

Let σ_i be a left-invariant coframe on S^3 normalized by

$$d\sigma_1 = -\sigma_2 \wedge \sigma_3, \text{quad}\sigma_2 = -\sigma_3 \wedge \sigma_1, \text{quad}\sigma_3 = -\sigma_1 \wedge \sigma_2, \text{quad} \int_{S^3} \sigma_1 \wedge \sigma_2 \wedge \sigma_3 = 16\pi^2. \quad (9)$$

We use metric signature $(-, +, +, +)$. The curvature convention is fixed operationally by

$$\delta \int \sqrt{-g} C^2 = 4 \int \sqrt{-g} B_{\mu\nu} \delta g^{\mu\nu}, \quad (10)$$

which, with the matter stress convention below, gives $\alpha_B B_{\mu\nu} = T_{\mu\nu}$. Set

$$g = -dt^2 + a^2(\sigma_1^2 + \sigma_2^2) + c^2\sigma_3^2, \quad q = \frac{c^2}{a^2}. \quad (11)$$

The matter fields are two real conformal scalars with positive target metric, written in polar form as

$$T_1 = \rho \cos(\omega t), \quad T_2 = \rho \sin(\omega t). \quad (12)$$

Their phase $\theta = \omega t \pmod{2\pi}$ obeys $g^{-1}(d\theta, d\theta) = -\omega^2 < 0$ and is therefore an everywhere timelike S^1 -valued clock candidate. The temporal-diffeomorphism/Weyl incidence determinant is $\rho^2\omega$, so the clock slice is regular on the branch below.

Our action convention is

$$S = \int_M \sqrt{-g} \left[\frac{\alpha_B}{8} C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} - \frac{1}{2} \sum_{A=1}^2 \nabla_\mu T_A \nabla^\mu T_A - \frac{R}{12} \rho^2 - \frac{\lambda}{4} \rho^4 \right]. \quad (13)$$

With $T_{\mu\nu} = -2\delta S_m/\delta g^{\mu\nu}$, its metric equation is

$$\alpha_B B_{\mu\nu} = T_{\mu\nu}. \quad (14)$$

Proposition 2.1 (Exact positive Berger family; P09-C1, P09-C2). *For*

$$\frac{5 - \sqrt{21}}{2} < q < \frac{1}{4} \quad (15)$$

there is an exact solution of the scalar and metric equations with

$$\rho^2 = \frac{2\alpha_B(1-4q)}{a^2}, \quad \omega^2 = \frac{q}{4a^2(1-4q)}, \quad \lambda = -\frac{q^2 - 5q + 1}{6\alpha_B(1-4q)^2} > 0. \quad (16)$$

The scalar kinetic metric is positive, the quartic potential is bounded below, and the exact stress obeys the pointwise dominant-energy inequalities $0 \leq p_h, p_v \leq \varepsilon$ in the orthonormal frame.

Proof. In the orthonormal Berger frame the independent Bach components are

$$B_{00} = \frac{(a^2 - c^2)^2}{6a^8}, \quad (17)$$

$$B_{11} = B_{22} = \frac{(a^2 - c^2)(a^2 - 3c^2)}{6a^8}, \quad (18)$$

$$B_{33} = \frac{(a^2 - c^2)(5c^2 - a^2)}{6a^8}. \quad (18)$$

Substitution of (12) into the scalar equation and the three independent components of (14) gives (16). The interval (15) makes ρ^2 , ω^2 and λ positive. Direct substitution gives

$$\begin{aligned} \varepsilon &= \frac{\rho^2(1-q)^2}{12a^2(1-4q)}, \\ p_h &= \frac{\rho^2(1-q)(1-3q)}{12a^2(1-4q)}, \quad p_v = \frac{\rho^2(1-q)(5q-1)}{12a^2(1-4q)}, \end{aligned} \quad (19)$$

from which $0 \leq p_h, p_v \leq \varepsilon$ follows on the stated interval. The coefficientwise verification also checks the lapse, horizontal-scale and vertical-scale variations of the Weyl-square action. $\square$

These properties define the word “healthy” in this paper: standard-sign target kinetic form, nonzero amplitude, timelike phase gradient, positive bounded-below potential on the solution, and the displayed pointwise dominant-energy inequalities. They are properties of this exact solution, not a claim that the improved conformal-scalar stress satisfies an energy condition on arbitrary configurations [16]. The inequalities are pointwise and on shell in the displayed Weyl representative; they are not asserted to be invariant under changing that representative.

Remark 2.2 (S^1 clock and local real lifts). Because $\rho > 0$, the map $(T_1, T_2)/\rho$ is globally S^1 -valued and has period $2\pi/\omega$ along the D orbit. A real clock coordinate exists only after choosing a local lift of θ on an interval shorter than one period. The slice $\theta = \theta_0$ therefore meets a complete orbit periodically. We prove transversality of this local clock slice, not a global one-intersection deparametrization and not absence of Gribov repetitions.

The rational point

$$a = 1, \quad q = \frac{9}{40}, \quad \alpha_B = 5, \quad \rho = 1, \quad \omega = \frac{3}{4}, \quad \lambda = \frac{119}{480} \quad (20)$$

is used for the exact all-row BV computations. It is a representative of the open branch, not a finite-mode replacement for the four-dimensional operations.

3 Nonzero clock momentum and vanishing tangent charge

The matter action has an internal $O(2)$ rotation

$$R(T_1, T_2) = (-T_2, T_1) \quad (21)$$

with current

$$j_R^\mu = T_1 \nabla^\mu T_2 - T_2 \nabla^\mu T_1. \quad (22)$$

For the Maurer–Cartan volume convention $\int_{S^3} \sigma_1 \wedge \sigma_2 \wedge \sigma_3 = 16\pi^2$, the background charge is

$$Q_R = 16\pi^2 \alpha_B q \sqrt{1 - 4q} > 0. \quad (23)$$

This is the canonical momentum conjugate to the phase θ . Thus the clock evolves and carries a genuine momentum.

The cylinder generator $D = \partial_t$ acts trivially on the stationary metric and rotates the scalar pair, as already distinguished from R and K in (3):

$$\mathcal{L}_D g = 0, \quad \mathcal{L}_D T = \omega R T. \quad (24)$$

The action-derived covariant presymplectic form then gives the exact identity

$$\Omega_{\text{total}}(\delta, \mathcal{L}_D) = \omega \delta Q_R. \quad (25)$$

The pure-Weyl cross term vanishes because $\mathcal{L}_D g = 0$; the matter term is the variation of the Noether charge (23). The Iyer–Wald improvement is an exact spatial divergence, whose integral vanishes because the Cauchy surface S^3 is closed. This establishes P09–C3, but by itself it does not decide whether D is gauge. The decision depends on the allowed tangent space.

3.1 The fixed-coupling slice and the lapse normalization

To define that tangent space without gauge-fixing away its constraint, retain the homogeneous lapse and use the common Weyl scale to set $a(t) = 1$ only after variation:

$$ds^2 = -N(t)^2 dt^2 + \sigma_1^2 + \sigma_2^2 + c(t)^2 \sigma_3^2, \quad T = \rho(t)(\cos \theta(t), \sin \theta(t)). \quad (26)$$

With $V_0 = 16\pi^2$, the reduced action is

$$\frac{S_{\text{red}}}{V_0} = \int dt N c \left[\frac{\alpha_B}{8} C^2 + \frac{\dot{\rho}^2 + \rho^2 \dot{\theta}^2}{2N^2} - \frac{R\rho^2}{12} - \frac{\lambda\rho^4}{4} \right]. \quad (27)$$

For auditability, the curvature scalars before the static specialization are

$$R = -\frac{c^3 N^3 - 4cN^3 + 4\dot{c}\dot{N} - 4N\ddot{c}}{2cN^3}, \quad (28)$$

$$C^2 = \frac{4A_- A_+}{3c^2 N^6}, \quad A_{\pm} = c^3 N^3 - cN^3 + \dot{c}\dot{N} - N\ddot{c} \pm 3c\dot{c}N^2. \quad (29)$$

Define the normalized lapse equation

$$E_N := \frac{1}{V_0} \frac{\delta S_{\text{red}}}{\delta N}. \quad (30)$$

Equations (27)–(30) fix the normalization in the linear identity below.

The open background family is a family across theories, not a modulus of one fixed theory. Eliminating the background variables gives

$$F(q; \alpha_B \lambda) = q^2 - 5q + 1 + 6\alpha_B \lambda (1 - 4q)^2 = 0, \quad (31)$$

and $\partial_q F$ after substituting the branch value of $\alpha_B \lambda$ is $3(6q - 1)/(4q - 1)$, which never vanishes on (15). Schematically,

$$\{\bar{\Phi}(q), \alpha_B, \lambda(q)\}_{q \in I} \longrightarrow \{(\alpha_B, \lambda)\}, \quad T_{\bar{\Phi}} \mathcal{S}_{\alpha_B, \lambda} = \ker D\mathcal{E}_{\bar{\Phi}} \cap \ker d\alpha_B \cap \ker d\lambda. \quad (32)$$

Thus $\delta N, \delta c, \delta \rho, \delta \theta$ are varied in deriving the equations, but an allowed tangent has $\delta \alpha_B = \delta \lambda = 0$ and must satisfy every linearized equation, including $\delta E_N = 0$.

Proposition 3.1 (Fixed-coupling rigidity of the Berger clock momentum; P09–C4). *On the smooth fixed- α_B , fixed- λ linearized solution space about the positive Berger clock background,*

$$\delta Q_R = 0. \quad (33)$$

Hence (7) holds as its covariant-phase-space corollary.

Proof. Direct variation of (27), followed by restriction to the background branch, gives

$$\delta E_N = -\frac{\alpha_B q^{3/2}}{2} \frac{\delta Q_R}{Q_R}. \quad (34)$$

Here

$$\frac{\delta Q_R}{Q_R} = \frac{\delta c}{\sqrt{q}} + 2\frac{\delta \rho}{\rho} + \frac{\delta \dot{\theta}}{\omega} - \delta N \quad (35)$$

at $a = N = 1$. The coefficient in (34) is nonzero throughout (15), so every homogeneous allowed tangent has $\delta Q_R = 0$. At the rational fixture the same row is

$$\delta E_N = \frac{27\sqrt{10}}{320} \delta N - \frac{9}{16} \delta c - \frac{9\sqrt{10}}{80} \delta \dot{\theta} - \frac{27\sqrt{10}}{160} \delta \rho = -\frac{27\sqrt{10}}{320} \frac{\delta Q_R}{Q_R}. \quad (36)$$

By Lemma 3.2, the compact spatial isometry group preserves the background and the linearized equations, while δQ_R is an invariant linear functional. If a smooth inhomogeneous solution tangent had nonzero δQ_R , averaging it over the compact group would produce a homogeneous solution tangent with the same nonzero value, contradicting (34). Thus (33) holds on the complete declared smooth tangent space. Equation (25) gives the result. $\square$

Lemma 3.2 (Compact averaging). *Let $G = SU(2)_L \times U(1)_R$ be the compact spatial isometry group of the Berger background and let $\text{Av}_G \delta\Phi = \int_G g \cdot \delta\Phi d\mu_G(g)$. Then Av_G is a continuous map in the smooth Fréchet topology, commutes with the fixed-coupling linearized equations, and obeys*

$$\delta Q_R(\text{Av}_G \delta\Phi) = \delta Q_R(\delta\Phi). \quad (37)$$

Proof. Naturality of the equations and G -invariance of $\bar{\Phi}$ give $D\mathcal{E}_{\bar{\Phi}}(g \cdot \delta\Phi) = g \cdot D\mathcal{E}_{\bar{\Phi}}(\delta\Phi)$; Haar integration therefore preserves the kernel and all linearized constraints. The internal charge is the integral over the closed spatial slice of a G -invariant scalar density, so its differential is a G -invariant continuous linear functional. Haar normalization then gives the displayed equality. Residual-isometry zero modes are retained by the average; none is removed by a spectral projector. $\square$

The compact averaging step is the matter-clock analogue of the familiar linearization-stability constraints on compact Cauchy surfaces [9, 10]; here the exact lapse identity (34) supplies the decisive coefficient.

Remark 3.3 (What is fixed). The interval (15) parametrizes theories through the product $\alpha_B \lambda$; it is not a modulus inside one fixed theory. At fixed couplings the stationary branch does not furnish a freely variable q or Q_R . There is therefore no contradiction between $Q_R > 0$ and $\delta Q_R = 0$ on allowed tangents.

4 Gauge explanation and relational evolution

The phase θ changes along D , but a gauge transformation of the total system need not leave every partial observable fixed. Relational quantities compare another field to the clock. If X is a scalar quantity with $DX \neq 0$, then locally one may evaluate X on the slice $\theta = \theta_0$ or form combinations invariant under the simultaneous shift of X and θ . The background momentum (23) makes this clock slice nondegenerate, while Proposition 3.1 says that the infinitesimal cylinder motion is presymplectically null in the compact fixed-coupling tangent space.

This paper proves that θ is a regular, backreacting clock candidate and that D is null in the specified linearized presymplectic sense. Its homological theorem concerns the distinct background stabilizer K . It does *not* construct a complete observable in the sense of Rovelli–Dittrich [4, 5]: no global gauge-invariant map $X \mapsto X|_{\theta=\theta_0}$, uniqueness domain, or complete-observable Poisson algebra is asserted. The periodicity in Remark 2.2 is one reason to keep that distinction explicit. Relational Maxwell redshift fixtures in the repository are follow-up applications and are not dependencies of Theorems 1.2 or 1.4.

This result is sector dependent. A boundary Hamiltonian, nonzero flux, or a phase space allowing $\delta Q_R \neq 0$ would change the classification of D . Nothing here promotes the compact Berger verdict to asymptotically flat, de Sitter, anti-de Sitter, or bounded regions.

5 The complete causal BV complex

Let $(\mathcal{C}_{54}, q_1)$ denote the gauge-fixed classical BV complex at the rational point (20). Its row degrees have multiplicities

$$5 \text{ in degree } -1, \quad 22 \text{ in degree } 0, \quad 22 \text{ in degree } 1, \quad 5 \text{ in degree } 2. \quad (38)$$

Table 1 summarizes the component implementation. Here τ is the clock-adapted time ghost, σ the Weyl row, and (R, Θ) the dressed amplitude and phase rows.

Degree	Bundle rows	Minimal	Nonminimal
-1	spatial diffeomorphism ghosts c_i , temporal ghost τ , Weyl ghost σ	5	0
0	dressed metric $\hat{h}_{\mu\nu}$, R, Θ ; multipliers b_a ; antighost duals $\bar{c}_a^*$	12	10
1	$\hat{h}_{\mu\nu}^*, R^*, \Theta^*$; multiplier duals b_a^* ; antighosts $\bar{c}_a$	12	10
2	ghost/identity antifields c_i^*, τ^*, σ^*	5	0
Total		34	20

Table 1: The complete 54-component gauge-fixed presentation. The five nonminimal labels a comprise three spatial-diffeomorphism, one temporal and one Weyl row.

The invariant retained complex has four bundle rows

$$\Gamma(T^*S^3)[-1] \longrightarrow \Gamma(S^2T^*M) \longrightarrow \Gamma(S^2T^*M)[1] \longrightarrow \Gamma(T^*S^3)[2], \quad (39)$$

of component ranks 3, 10, 10, 3. Its unary blocks are the spatial gauge generator, the retained action Hessian and the negative formal-adjoint gauge identity. The remaining eight minimal clock/temporal/Weyl rows and twenty nonminimal rows form the 28-component contractible extension. Thus the number 54 is an all-row completeness audit, whereas the statement $\mathcal{C}_{54} \simeq \mathcal{C}_{26}$ is independent of the chosen component ordering.

The analytic core can be stated without row data. The retained ten-component metric Hessian is

$$A_{10} = \square_2^2 + V_2, \quad \text{ord } V_2 \leq 2, \quad (40)$$

where $\square_2$ is the normally hyperbolic rough tensor wave operator and V_2 has stationary smooth coefficients. It is the Schur compression of the companion system

$$C_{20} = \begin{pmatrix} \square_2 & -I_{10} \\ V_2 & \square_2 \end{pmatrix}. \quad (41)$$

On a finite time slab I , let

$$X_s(I) = (C^0(I; H^{s+1}) \cap C^1(I; H^s)) \oplus (C^0(I; H^s) \cap C^1(I; H^{s-1})), \quad (42)$$

$$Y_s(I) = L^1(I; H^s) \oplus L^1(I; H^{s-1}), \quad (43)$$

with Sobolev spaces on S^3 valued in symmetric tensors. Writing the free wave companion as C_0 and its local coupling as N , Volterra ordering gives distinct solution- and source-space resolvents

$$R_{\text{sol},\pm} = (I + G_{0,\pm}N)^{-1}, \quad R_{\text{src},\pm} = (I + NG_{0,\pm})^{-1}, \quad (44)$$

and

$$G_{C,\pm} = R_{\text{sol},\pm}G_{0,\pm} = G_{0,\pm}R_{\text{src},\pm}. \quad (45)$$

The finite-slab estimates

$$\|(G_{0,\pm}N)^n\|_{X_s \rightarrow X_s}, \quad \|(NG_{0,\pm})^n\|_{Y_s \rightarrow Y_s} \leq \frac{C_{s,I}^n}{n!} \quad (46)$$

give absolute convergence, both inverse identities and causal support. Compression through the local companion graph yields $G_{A,\pm}$ for (40); uniqueness globalizes the finite-slab construction. The typed adjoint statement is

$$(G_{A,+})^\sharp = G_{A^\sharp,-}, \quad (47)$$

without assuming $A = A^\sharp$. The spatial gauge and identity rows use the corresponding wave companions and complementary-degree adjoints. These blocks assemble the retained chain homotopies below.

A gauge-fermion canonical shear preserves the frozen odd Darboux pairing and gives a support-local cyclic contraction

$$\pi_{\text{cl}}\iota_{\text{cl}} = \text{id}, \quad \text{id} - \iota_{\text{cl}}\pi_{\text{cl}} = q_1 S_{\text{cl}} + S_{\text{cl}} q_1 \quad (48)$$

onto the retained complex (39). This is claim P09-C5.

For compactly supported sources, the retained complex has advanced and retarded degree-(-1) chain contractions

$$\Lambda_{26,\pm} : \Gamma_c(\mathcal{C}_{26}) \longrightarrow \Gamma_{J^\pm}(\mathcal{C}_{26})[-1]. \quad (49)$$

Lifting them through (48) gives

$$\Lambda_{54,\pm} = S_{\text{cl}} + \iota_{\text{cl}}\Lambda_{26,\pm}\pi_{\text{cl}}, \quad (50)$$

and therefore

$$q_1\Lambda_{54,\pm} + \Lambda_{54,\pm}q_1 = \text{id}_{54}, \quad \text{supp}(\Lambda_{54,\pm}f) \subseteq J^\pm(\text{supp } f). \quad (51)$$

Indeed,

$$\begin{aligned} [q_1, \Lambda_{54,\pm}] &= [q_1, S_{\text{cl}}] + \iota_{\text{cl}}[q_1, \Lambda_{26,\pm}]\pi_{\text{cl}} \\ &= (1 - \iota_{\text{cl}}\pi_{\text{cl}}) + \iota_{\text{cl}}\pi_{\text{cl}} = 1. \end{aligned} \quad (52)$$

All maps in the retract are differential or pointwise algebraic. No inverse Laplacian, inverse curl, transverse-traceless projector, or mode projector is used. The complementary-degree adjoint relation and stationary-generator equivariance also lift. By Proposition 1.1, that generator is K . These statements comprise P09-C6.

The cyclic higher primitives below instead map compactly supported inputs to sections supported in the two-sided causal hull $J = J^+ \cup J^-$. They are neither the causal difference $\Lambda_+ - \Lambda_-$ nor separate advanced/retarded higher operations.

6 Action-derived interaction tensors

For a fluctuation ϕ about $\bar{\Phi}$, expand the classical BV vector field in the suspended graded-symmetric factorial convention,

$$Q_{\bar{\Phi}}(\phi) = q_1(\phi) + \frac{1}{2!}q_2(\phi, \phi) + \frac{1}{3!}q_3(\phi, \phi, \phi) + O(\phi^4). \quad (53)$$

Thus q_n has n inputs, and q_3 is the fourth action derivative paired with one output. If $\langle -, - \rangle_{\text{BV}}$ denotes the degree-one odd Darboux pairing, the operations are defined invariantly by the polarized identity

$$\langle x_0, q_n(x_1, \dots, x_n) \rangle_{\text{BV}} = \frac{\partial^{n+1}}{\partial t_0 \dots \partial t_n} S_{\text{BV}} \left[\bar{\Phi} + \sum_{j=0}^n t_j x_j \right] \Big|_{t=0}, \quad (54)$$

with the displayed ordering interpreted in the frozen suspended Koszul convention. The local clock dressing and gauge-fermion shear are then applied as BV-canonical transformations to every input, output and dual row.

The proof pipeline is therefore

$$S_{\text{BV}} \xrightarrow{\text{mixed variations}} (q_1, q_2, q_3) \xrightarrow{\text{Cartan sources}} A_K^{(1)}, A_K^{(2)}, A_K^{(3)} \xrightarrow{\text{Green/Hom contraction}} \iota_K^{(1)}, \iota_K^{(2)}, \iota_K^{(3)}. \quad (55)$$

The operations q_2 and q_3 are obtained by differentiating the covariant Weyl-clock master action on arbitrary four-dimensional jets and transporting the result through the same clock and gauge-fermion canonical maps used at arity one. They are not fits to residual modes or a harmonic cutoff.

The arity-two and arity-three parts of $Q^2 = 0$ are

$$[q_1, q_2] = 0, \quad (56)$$

$$[q_1, q_3] + \frac{1}{2}[q_2, q_2] = 0. \quad (57)$$

The common action origin supplies cyclicity with respect to the odd Darboux pairing. The stationary invariant-frame generator $K = e_0$ acts linearly, so its higher Taylor components vanish through this order,

$$L_K^{(2)} = L_K^{(3)} = 0, \quad (58)$$

and exact coefficientwise verification gives

$$[K, q_2] = [K, q_3] = 0. \quad (59)$$

Claims P09-C7 and P09-C8 bind these statements to the portable all-row coefficient exports, with the legacy D label interpreted by Proposition 1.1.

The abstract L_∞ relations follow from the classical master equation; the large PBW calculation verifies that the concrete curved coefficients, canonical transports and signs instantiate those relations without a dropped row. As a cross-producer check, an independently written homogeneous reduced action gives eight fourth derivatives in the $(\widehat{h}_{00}, R)$ sector. They agree exactly with all sixteen ordered zero-jet coefficients in the published $\widehat{h}_{00}^*$ and R^* rows. This probes both the lapse and dressed radial/Weyl directions without importing the q_3 producer. It is deliberately not described as a complete second derivation of the payload.

Arity three is the first test involving both q_3 and the composite $[q_2, q_2]$ channel. It is not the end of the classical expansion: inverse metrics, curvature and canonical field redefinitions generally generate nonzero q_n for $n \geq 4$ even though the bare scalar potential is quartic.

7 Causal K -Cartan contraction through arity three

Write

$$\iota_K = \iota_K^{(1)} + \iota_K^{(2)} + \iota_K^{(3)} + \cdots \quad (60)$$

for a degree-(-1) Cartan homotopy. At unary order,

$$[q_1, \iota_K^{(1)}] = K. \quad (61)$$

A one-sided raw solution follows from (51); averaging it with its convention-correct cyclic adjoint yields a cyclic unary primitive with support in the two-sided causal hull.

For clarity, the chain contraction is used on the multilinear Hom complex, not as an inverse of a coefficient matrix. If $\delta = [q_1, -]$ and A is a homogeneous multilinear operator, set

$$\mathfrak{H}_\pm(A) := \Lambda_{54, \pm} \circ A. \quad (62)$$

The graded derivation rule and $[q_1, \Lambda_{54, \pm}] = 1$ give

$$\delta \mathfrak{H}_\pm(A) + \mathfrak{H}_\pm \delta(A) = A. \quad (63)$$

This identity includes the q_1 action on every input slot; no input term is discarded. It is the step that turns a q_1 -closed Cartan source into a raw causal primitive.

Proposition 7.1 (Conditional finite-arity Cartan recursion). *Let $(\mathcal{C}, Q, \langle -, - \rangle)$ be a cyclic formal L_∞ algebra, let a background-fixing unary generator K satisfy $[Q, K] = 0$, and suppose q_1 admits cyclically compatible causal chain contractions. If the Taylor operations and K -equivariance are defined through arity n , then the Cartan equation can be solved cyclically through arity n , with each component supported in the two-sided causal hull of its inputs.*

Proof. Assume the equation has been solved below arity $m \leq n$ and denote by $A_K^{(m)}$ the known arity- m source. The truncated identity $[Q, [Q, \iota_K] - L_K] = 0$, using $Q^2 = 0$ and $[Q, L_K] = 0$, implies $[q_1, A_K^{(m)}] = 0$. Equation (62) gives a raw causal primitive $-\mathfrak{H}_+(A_K^{(m)})$. Pairing its output and applying the finite cyclic Reynolds projector preserves the homotopy equation because q_1 and the source are cyclic. Induction proves the statement. This is a formal finite-arity result; it asserts neither convergence of the Taylor series nor the existence of operations not supplied by the theory. $\square$

Thus the conceptual obstruction vanishes conditionally at every finite arity. The model-specific computation through arity three is still needed to certify the curved coefficients, canonical transports, K -equivariance and all-row cyclic signs. Operations and equivariance beyond q_3 have not been exported, so they are not promoted here.

At binary order define

$$A_K^{(2)} = [q_2, \iota_K^{(1)}]. \quad (64)$$

Equations (56) and (59) make $A_K^{(2)}$ q_1 -closed. A raw causal primitive is cyclicized with $\text{Cyc}_3 = (1 + \tau + \tau^2)/3$, giving

$$[q_1, \iota_K^{(2)}] = -[q_2, \iota_K^{(1)}]. \quad (65)$$

This is P09-C9.

The ternary source is

$$A_K^{(3)} = [q_3, \iota_K^{(1)}] + [q_2, \iota_K^{(2)}] - L_K^{(3)}. \quad (66)$$

Lemma 7.2 (Closure of the ternary source). *The complete arbitrary-input source (66) obeys*

$$[q_1, A_K^{(3)}] = 0. \quad (67)$$

Proof. Let $\delta = [q_1, \cdot]$. Using (57), (61) and (65), the only nonzero formal channels in $\delta A_K^{(3)}$ are

$$-\frac{1}{2}[[q_2, q_2], \iota_K^{(1)}] + [q_2, [q_2, \iota_K^{(1)}]] - [q_3, K]. \quad (68)$$

Graded Jacobi gives

$$[q_2, [q_2, \iota_K^{(1)}]] = \frac{1}{2}[[q_2, q_2], \iota_K^{(1)}], \quad (69)$$

and $[K, q_3] = 0$ annihilates the last term. Thus the normalized coefficients are $-1/2 + 1/2 = 0$ and 0 in the two independent channels. $\square$

Because cyclic coderivations are closed under their graded bracket, $A_K^{(3)}$ is cyclic. The causal chain contraction supplies a raw primitive

$$R^{(3)} = -\mathfrak{H}_+(A_K^{(3)}) = -\Lambda_{54,+} \circ A_K^{(3)}, \quad \delta R^{(3)} = -A_K^{(3)}. \quad (70)$$

Pair the output with the odd Darboux form before cyclicizing. The resulting four-linear tensor carries the standard Koszul C_4 action and the exact Reynolds projector

$$\text{Cyc}_4 = \frac{1}{4}(1 + \tau + \tau^2 + \tau^3). \quad (71)$$

Since $\tau^4 = 1$, $\text{Cyc}_4^2 = \text{Cyc}_4$. Cyclicity of q_1 makes Cyc_4 commute with δ , while cyclicity of the source gives $\text{Cyc}_4 A_K^{(3)} = A_K^{(3)}$. Therefore

$$\iota_{K,\text{cyc}}^{(3)} = \text{Cyc}_4 R^{(3)}, \quad [q_1, \iota_{K,\text{cyc}}^{(3)}] = -A_K^{(3)}. \quad (72)$$

This proves the ternary part of Theorem 1.4 and P09-C10.

The tensorized C_4 convention matters. The actual 54-row pairing has 27 negatively oriented dual slots. Its duality involution, degree-one pairing and odd skew orientation are checked on every row. The C_4 group law is exhausted over all actual degree classes, covering 978,736 admissible degree-zero row quartets with no defect.

7.1 Support statement

The Taylor operations and pairing are support-local. Green contractions and their cyclic adjoints are causal. Consequently

$$\text{supp}(\iota_{K,\text{cyc}}^{(3)}(f, g, h)) \subseteq J(\text{supp } f \cup \text{supp } g \cup \text{supp } h). \quad (73)$$

The cyclic primitive is not asserted to be separately retarded or advanced. The one-sided chain contractions (51) retain that stronger property; cyclic averaging generally does not.

8 Interpretation, failures and open directions

The primary geometric result is the rigidity statement $Q_R > 0$ but $dQ_R|_{\mathcal{Z}_{\Phi}^{\alpha_B, \lambda}} = 0$. Presymplectic nullity of D is its covariant-phase-space corollary. The all-row BV calculation is an adjacent theorem about the background-fixing helical generator K . Proposition 1.1 is the precise bridge: it proves both that the two generators are related and that they must not be identified in the based Taylor complex. No derived symplectic equivalence between an affine D -Cartan complex and the certified K -Cartan complex is asserted.

The following conclusions are not drawn.

1. Proposition 7.1 gives a conditional formal finite-arity recursion, but operations and K -equivariance from arity four onward have not been exported. No convergent all-orders contraction is claimed.
2. No affine D -Cartan homotopy, including its zeroth-arity equation, is constructed.
3. No constant-rank nonlinear null foliation, integrated gauge quotient, global deparametrization or complete relational observable is constructed.
4. No Hadamard two-point function, microlocal spectrum condition, renormalized composite, anomaly cancellation or quantum master equation is constructed.
5. No positive graviton Hilbert space or interacting probability theorem follows from the positive clock matter.

6. No noncompact or bounded phase space is covered. There D may acquire a boundary charge or flux and cease to be presymplectically null.
7. Relational-redshift fixtures are applications, not part of the main theorem unless their independent propagation and backreaction gates close.
8. A separate exact **LOCAL-ALGEBRAIC/REDUCED-MODE** observer calculation tests the direct temporal A - K repair in an overcomplete action module: all 932 IBP-closed order-three scalar-density actions per emitter, both reflection parities and differentiated $\chi/K/A$ placements, plus the electric and magnetic clock-profile lines. The direct τ^* projection is repairable: its action and source-augmented ranks are both 934. Full covariance nevertheless fails on the joint τ^*, A_0^+ projection, where the ranks are $934 < 935$ and the first exact quotient coefficient is

$$A_0^+ \leftarrow (\tau, e_0 e_1 K_{0,01}) = -3g_0 h_0.$$

Freeing the inherited emitter Diff-BV normalization changes these ranks to $935 < 936$, so the obstruction is not a normalization artifact. The complete bounded existing-row Maxwell/emitter-antifield module has now also been classified. Per emitter it contains 2,048 exact local-action Hessians: all invariant orders zero through two and both $A^+/\tau/K$ and cyclic $K^+/\tau/A$ transforms of all 932 inherited IBP-closed order-three actions. These add quotient rank 1,679, but the full ranks remain $2613 < 2614$, with the same 42-coordinate quotient and first coefficient $-3g_b h_b$. Hence the first unexcluded representation requires a genuinely new q_1 -preimage of A^+ , together with its Maxwell ghost-antifield and signed cotangent completion. That first new representation has now been classified in a separate exact calculation. The minimal connected Berger- $U(1)$ choice is the four-row curl doublet

$$q_1 B_{01} = e_1 A_0^+ - e_0 A_1^+, \quad q_1 B_{02} = e_2 A_0^+ - e_0 A_2^+,$$

with signed cotangent partners. Its new nilpotency compositions and unary cyclicity defect vanish exactly. The complete declared local first-jet class has four order-zero and twenty-four order-one invariant action Hessians per emitter, but adds only four quotient directions. The ranks therefore remain $2617 < 2618$, with the same 42-coordinate representative. Modulo the full image, the source-isolated obstruction is the weight-zero $U(1)$ class. This is a no-go only for that declared four-row first-jet class; the least unexcluded enlargement is its complete PBW/IBP order-two action prolongation. That prolongation has now also been evaluated. Its PBW-filtered $U(1)$ generator has rank 404 on 528 raw $g_b h_b$ order-one/two monomials and hence a 124-dimensional kernel. Together with four independent $g_b h'_b$ directions, this gives 128 exact local actions and Hessians per emitter. They add quotient rank 28, but the full ranks are still $2641 < 2642$, and the same 42-coordinate source representative survives. Thus the no-go now covers the declared four-row filtered class through total action order two; it does not cover any all-finite-order differential-symbol module or larger mixed bundle. The next outcome-safe gate is therefore not another isolated jet order. It is a finite presentation, with PBW/IBP syzygies and a certified filtered lift, over the connected Berger- $U(1)$ differential-symbol ring. The audited exact-math substrate currently lacks that module layer, so the all-jet membership question is recorded as an open shortfall rather than inferred from finite-order rank stabilization. In particular, no irreducible all-jet cokernel or minimal excluded enlargement has yet been certified. Since the arity-two admissible locus is still empty, no common-action quartic q_3 is promoted; the previous $-4g_0 h_0$ arity-three coefficient remains only a diagnostic. In particular, although the ambient Maxwell/Berger- $U(1)$ quartic family has twelve directions and a rank-twelve q_3 parameter map, it has no admissible nonlinear theory point after the

preceding master identities. The moduli-to-detector polynomial and its calibration rank are therefore *not applicable on an empty domain*, not a zero map or a completion-independent prediction. No nonlinear observer morphism, redshift or detector-rank theorem is claimed.

An orthogonal apparatus construction can proceed without changing that all-jet boundary. A minimal real first-order material action now carries two detector rod-orientation doublets, two polarization doublets, two persistent memory pairs and two emitter-phase doublets, together with all odd cotangents. Its quadratic part derives the unary differential and the cubic $-\lambda_a p_a \cdot F_a$ derives the detector binary operation. The resulting action identities, real structure, simultaneous Berger- $U(1)$ covariance and clock-transport principal symbols close exactly through arity two. On the probe branch the induced response is the previous triangular matrix with determinant $\kappa_0 \kappa_1 \neq 0$, so the leading rank-two record survives. The new binary coordinates all contain an apparatus input and therefore have zero support intersection with the pure-old 42-coordinate finite second-jet source representative. This preserves, rather than answers, the separate all-jet question. Restriction of the new response to the second-order cone and a relational emitted/received phase comparison remain open.

There is a further linear reduction prerequisite. The material parent currently specifies its action, field/parity list, aggregate odd-pairing rank and transport symbols, but not a row-level combined differential with the Berger gravity-clock-Maxwell carrier. Since the existing 108-row carrier already contains rods, memories and emitters, a concatenation based on matching names would double count rather than define a complex. A machine-readable crosswalk contract now requires an explicit base-carrier choice, row embeddings, sparse q_1 , row-level pairing and grading, real and K_{Berger} actions, detector-smearing chain map and separate generic-frequency/zero-mode sectors. Until it is instantiated, neither the physical apparatus cohomology nor the descended memory classes and rank-two response are claimed.

Even after that linear reduction interface is supplied, a distinct same-background second-order receiver is required. The parent does not currently export the quadratic source on the three symmetric preparation pairs

$$(u_0, u_0), \quad (u_0, u_1), \quad (u_1, u_1),$$

the complete set of their Berger output blocks, the stabilizer moment-map/Taub projections, or the complementary nonzero-shell adjoint cokernels and resonant pairings. The general finite-harmonic tangent-cone theorem gives the image/cokernel criterion but no map from its compact-product fixture to this Berger carrier. A machine-readable receiver contract therefore keeps bounded/quasiperiodic, smooth secular and causal/retarded correction classes separate and requires an exact amplitude ideal in each class before restricting the detector matrix. Until the requested same-background receiver is delivered, the leading rank-two result remains only a probe-branch linear statement: no individual mode, balanced combination, nonlinear response rank, gauge-reduced memory or relational redshift is inferred.

The combined-unary crosswalk has now been tested in the declared typed constant-identification class. Identifying the two memories, two multipliers and their cotangents gives a 156-row candidate, while the material orientation, polarization and phase rows remain distinct from the global scalar rods and massive two-form fields. The candidate fails at the required background-preserving linear K_{Berger} interface. The six global rods have time-translation closure rank eight and no constant internal six-by-six completion. Nor can the first-order transported material rows supply the missing directions: coefficientwise comparison of $sA_4B = Cs^2$, with $\det A_4 = 1$, has rank 24 on 24 constant mixing variables and zero nullity per detector. The minimal pairing-preserving repair is two new global rods and two cyclic cotangents, followed by recomputation of the co-rotating background,

rod stress, Φ_2 , the 112-row unary and its quotient. Thus the prospective 160-row identified union is a next construction, not a current physical complex.

That minimal repair has now been carried through its first later interface. The normalized sine-phase partners of the first global rod at each detector, with their signed cyclic cotangents, complete the centered background K_{Berger} -orbit from rank six to rank eight. The eight-rod stress is Noether closed in both retained frequency blocks, and exact sparse primitives solve the recomputed $\Phi_2^{(8)}$ background equation. Nevertheless the same scalar action produces a nonzero mixed metric-to-new-rod-cotangent principal symbol cs at both detector events. Consequently the canonical inclusion of the certified 108-row unary has two chain defects. A local differential correction would have to satisfy $-s^2 P(s) = cs$ with $c \neq 0$, while the only formal solution $P = -c/s$ has differential order minus one. The four-row class is therefore obstructed as a local extension of the old complex. This result does not certify a complete 112-row unary, the prospective 160-row apparatus union, a support-controlled nonlocal comparison, or any reduced detector, memory, redshift or tangent-cone observable.

Treating the same 112 rows as a replacement theory rather than an extension does not rescue the certified diagonal rod action. In the exact rank-eight background basis, the forced generator A has symmetric part of rank four and diagonal entries $\pm 2 \cot(\sqrt{58}/24)$. Its canonical cotangent lift is $-A^T$, so the identity-kinetic scalar-wave Hessian has normalized principal K_{Berger} -commutator $-(A^T + A)$, also of rank four. Deleting either new pair leaves background rank seven, whereas replacing A by its skew part loses background closure with rank-four defect. The positive non-diagonal kinetic matrix $H = B^{-T}B^{-1}$ is an exact unexcluded repair, but it changes the action and therefore requires fresh stress, Φ_2 , unary and quotient calculations. No complete 112-row complex or downstream observable is promoted by this obstruction.

The smallest changed-action repair does pass the unary gate. Set $H = B^{-T}B^{-1}$, write the eight rods as $R = B\psi$, and transform their cotangents by $R^+ = B^{-T}\psi^+$. The mixed rod action and odd pairing then become eight canonical positive scalar BV pairs, while $J = B^{-1}AB$ is the standard skew complex structure. The recomputed stress is homogeneous and time independent; its Noether and retained cokernel projections vanish and it has an exact four-entry Φ_2 primitive. The resulting action-derived 112-row unary passes nilpotency, cyclicity, reality and both principal and lower-order K_{Berger} covariance. Its rod block has support-local retarded and advanced parents $G_{\text{scalar}}H^{-1}$, without asserting a complete off-shell metric-BV propagator. The unchanged leading detector map remains rank two before full 112-row cohomological reduction. The latter reduction and every apparatus q_2, q_3 , tangent-cone, memory and redshift statement remain separate gates.

At unary order the typed material-apparatus pushout is now complete. The 168-row direct sum of the replacement base and material parent has exactly eight independent semantic relations, identifying two memories, two multipliers and their cotangents, and hence a derived 160-row quotient. Every other material row remains distinct. The complete q_1 , full-rank odd pairing, real and K_{Berger} actions, both embeddings, quotient, support sectors and detector-smearing chain map pass exactly. The leading coordinate response remains rank two, but its 160-row cohomological descent and every q_2, q_3 statement remain unclaimed.

The physical reduction is nevertheless not yet defined by the exported machine data. The unary payload records its action-level composition and identity defects, but not a coefficient ring, sparse q_1 entries, support-sector chain groups or zero-mode matrices. Because the mixed-action Φ_2 changes non-rod unary rows, the older executable 108-row operator is not replacement cohomology. Thus no exact kernel, image, contraction, descended pairing, persistent-memory class or reduced detector rank is claimed before a sectorwise executable 160-row operator is exported.

The same missing input also stops the repaired-apparatus second-order receiver before any

mode-level obstruction can be claimed. Without the physical preparation, detector and memory classes supplied by an exact contraction, the combined quadratic source has no defined domain. The abstract finite-harmonic image/cokernel theorem does not manufacture a Berger carrier map. Accordingly the bounded/quasiperiodic, smooth secular and causal/retarded Berger tangent cones, the detector response restricted to them, its nonlinear rank and persistent relational memory are all `NO_CERTIFIED_MAP`. This retains the coordinate-level leading rank-two response only in its linear parent scope; it neither proves nor disproves nonlinear detectability of an individual or balanced preparation.

An attempted executable serialization of the unary pushout identifies the smaller missing interface. The replacement 112-row payload gives its row table, pairing and action-level Hessian identities but no normalized row-indexed q_1 coefficients. The 56-row material-parent payload gives row lists, its action and principal transport symbol, and a pairing rank, but no unary or odd-pairing entries. Hence an independent consumer cannot reconstruct the two base matrices or compare their 160-row quotient byte for byte. The executable old 108-row blocks are not substitutable because the positive-mixed Φ_2 and four added rod/cotangent rows change the operator. Separate content-addressed base-producer exports are required before the physical-reduction gate can be reopened.

The first base-producer repair is itself stopped by a missing variational interface. The old universal executable correction contains named local Φ_2 component-jet factors, while the positive-mixed producer exports only four retained-harmonic coefficients for its new primitive. No certified map evaluates those coefficients into the required local jets. Consequently the non-rod correction $D^3 S_{108}[\Phi_0](\Phi_2^{\text{mixed}}, \cdot, \cdot)$ is not machine-defined. This smaller crosswalk and evaluated derivative must land before the replacement 112-row matrix or the material-parent producer is attempted.

That retained-to-local crosswalk is now exact. In the oriented orthonormal Berger frame the positive-mixed primitive has constant diagonal components

$$\Phi_{2,00} = \frac{428}{567}, \quad \Phi_{2,11} = \Phi_{2,22} = -\frac{29}{21}, \quad \Phi_{2,33} = -\frac{6}{7}.$$

All 942 component-PBW jets consumed by the universal correction through order five are evaluated. The frame commutators are checked exactly and the Levi–Civita connection and covariant tensor jets are exported separately. The resulting 6171 dependent terms reduce to 20 normalized survivors, while an independent coarea variation gives $-\Phi_{2,00}/2 = -214/567$. This closes only the variational-input interface: the complete executable replacement unary, the material-parent unary, their quotient and every reduced or nonlinear observer claim remain fail-closed. The machine-readable receipt is `closed_universe_observers/certificates/BERGER_POSITIVE_MIXED_PHI2_LOCAL_COMPONENT_JET_EXPORT.json`.

The parallel material-parent serialization is more limited. Its canonical 56-row dictionary, full signed pairing, six internal D_K Hessians and two memory Hessians are executable, including their generic and zero-mode matrices. But the declared quadratic action also contains $-\lambda_a \bar{P}_a \cdot F_a$. Its four ordered nonzero mixed derivatives cannot be assigned to rows because the certified inputs provide no F_a -to-base-row and detector-profile interface. The full material-parent unary therefore remains fail-closed; see `closed_universe_observers/certificates/BERGER_MATERIAL_PARENT56_EXECUTABLE_UNARY_EXPORT_SHORTFALL.json`.

Closing the positive-mixed Φ_2 jets likewise does not yet serialize the replacement unary. The non-rod correction is now exact, but the replacement action is specified as the old six-rod action removed and an H -weighted eight-rod action inserted without row-indexed $D_g D_R S_{R,H}$, subtraction/addition, Diff–BV adjoint, support or zero-mode entries. The first remaining object is therefore the normalized

mixed metric-rod Hessian, not another Φ_2 component; see `closed_universe_observers/certificates/BERGER_REPLACEMENT112_EXECUTABLE_UNARY_AFTER_PHI2_MAP_SHORTFALL.json`.

The missing action interface has subsequently been exported directly from $S_{\text{nonrod}} - S_{R,I_6} + S_{R,H}$. The certificate `closed_universe_observers/certificates/BERGER_REPLACEMENT112_MIXED_METRIC_ROD_HESSIAN_INTERFACE.json` contains the row-indexed six-rod subtraction, the $H = B^{-T}B^{-1}$ -weighted eight-rod addition and their normalized PBW difference for the rod gauge pair, wave block, mixed metric-rod Hessian, its coefficient-aware formal transpose and the rod-induced metric Hessian. The signed odd adjoints, exact mixed K_{Berger} representation, support matrices and retained hyperbolic spatial zero mode pass independent reconstruction. This closes the interface only: the complete 112-row differential, its global nilpotency/cyclicity replay, cohomology and the 160-row consumer remain successor gates.

That successor replay is obstructed already in the mixed unary square. At the exact non-degenerate unit-circle fixture $((s_a, c_a) = (3/5, 4/5))$, $((s_u, c_u) = (5/13, 12/13))$, the coefficient $(q_{10}q_{00} + q_{00}q_{10})$ has 132 nonzero Fourier-mode entries in 28 matrix positions after the declared $(R = B\psi)$ and sphere quotient. The eight rod wave equations vanish, while the canonical first witness is $(h_{00}^* \leftarrow \sigma)$ at time mode -2 . The exact ledger is `closed_universe_observers/certificates/BERGER_REPLACEMENT112_EXECUTABLE_UNARY_MIXED_NILPOTENCY_OBSTRUCTION.json`. Consequently the assembled rows are a diagnostic obstruction rather than an executable chain complex, and no 160-row, cohomological or operational observer result follows from them.

The material readout interface has now been made explicit: $F_a(A) = Q_a[dA]$ on Maxwell rows 55–58 with the certified compact clock/rod profile. The two forward memory-cotangent blocks and two signed adjoints reproduce the four ordered derivatives of $-\lambda_a \bar{P}_a \cdot F_a$, all with action coefficient -1 . Gauge-chain and zero-mode compatibility follow from $d^2 = 0$. This closes the interface only; see `closed_universe_observers/certificates/BERGER_MATERIAL_PARENT56_BACKGROUND_READOUT_INTERFACE.json`.

The resulting material module is executable on its own 56-row carrier. Its 52 internal unary entries have vanishing square, formal cyclicity, real and K_{Berger} commutator defects; the signed pairing has rank 56, the generic Hessian rank is 28 and the $s = 0$ rank is 24. The material detector selection has rank two. The four readout derivatives remain a typed external base-to-parent interface rather than fictitious internal 56-row entries. The exact export is `closed_universe_observers/certificates/BERGER_MATERIAL_PARENT56_EXECUTABLE_UNARY_AFTER_READOUT_INTERFACE.json`. This certifies only the material half: the obstructed replacement unary still prevents the combined 160-row pushout and its reduced observer algebra.

The complete declared local action-Hessian repair does not alter this conclusion. Its 4,542 exact nilpotency equations have coefficient rank one and augmented rank two, with a nonzero canonical two-equation minor; imposing the pinned background leaves correction dimension zero. Hence the fresh post-repair apparatus pushout is nondefined before an identification relation or combined row count is formed. The dependent physical reduction is therefore not activated: there is no domain for kernels, images, π_{cl} , a descended pairing or a two-record detector rank. The exact dispositions are `closed_universe_observers/certificates/BERGER_REPAIRED112_APPARATUS_PUSHOUT_AFTER_132_DEFECT_DISPOSITION.json` and `closed_universe_observers/certificates/BERGER_REPAIRED112_PHYSICAL_REDUCTION_AND_DETECTOR_RANK_NOT_ACTIVATED.json`. The standalone material rank-two coordinate selection is not promoted to a combined physical observable.

The correct continuation is therefore not to advertise an unrestricted solution of the problem of time. It is to test whether the K -Cartan structure survives higher Taylor order, whether an affine D -Cartan extension exists, whether the S^1 clock supports well-defined complete observables on controlled domains, and whether the conclusion persists under renormalized Ward identities and physical boundaries.

9 Reproducibility and claim map

The companion file `09-relational-clocks-berger-d-cartan-computational-supplement.tex` contains the focused claim ledger, exact certificate paths, coefficient inventory, replay commands and freeze checklist. Its machine-readable source is `../d_quotient_classical/certificates/PAPER_09_BERGER_CLAIM_TABLE.json`. The verifier checks strict Draft 2020–12 schema validity, certificate hashes, required true and false flags, and occurrence of every claim identifier in the manuscript sources.

The publication-current superset is `09-relational-clocks-berger-d-cartan-claim-map.json`. It preserves P09-C1–P09-C10 and separately types the observer dispositions P09-O1–P09-O12. The latter include conditional theorems, carrier-scoped fixtures, local and action-integration layers, physical non-descent, open higher modes and nonactivation. They contain no claim classified as an operational observable. The prior ten-claim scientific theorem retains its frozen scope, while this publication-current edition is `DRAFT_ALLOWED` until the legacy source-binding verifier is formally superseded or repinned and the full observer rail is green. These are evidence gates, not new scientific or apparatus gates.

The expensive q_3 action expansion is not silently rerun by the paper build. Its content-addressed Tier-2 receipt records the derivation, canonical transport, row-bounded identity replay and independent portable consumer. Fast paper checks import that frozen artifact by hash. The theorem-freeze receipt records the isolated clean-tree replay, environment, compiled PDFs and nonlinear- and quantum-team scope reviews. It promotes no claim beyond the ten entries in the companion ledger.

9.1 Charged physical time and conditional event maps

There is a distinct relational construction when the clock translation is a charged physical symmetry rather than a gauge direction. Let (ψ, Q) be a surviving action–angle pair with $\{\psi, Q\} = 1$, Hamiltonian $H(Q)$, phase generator $R = X_Q$, and $K = D - H'(Q)R$. For a compact local receiver cocycle A of form degree $d - 1$ satisfying $sA + dB = 0$, the distributional current

$$\mathcal{O}_A(\tau) = \int_M \delta(\psi - \tau) d\psi \wedge A \quad (74)$$

is locally BV closed and depends only on the receiver class. It is covariant, not invariant by quotienting, under the charged global flows:

$$\{\mathcal{O}_A, Q\} = -\partial_\tau \mathcal{O}_A, \quad \{\mathcal{O}_A, H\} = -H'(Q)\partial_\tau \mathcal{O}_A. \quad (75)$$

The K bracket is the event map of $\mathcal{L}_K A$ and vanishes only for a declared K -basic receiver.

Primitive clock changes are $\psi' = \epsilon\psi + s$, $Q' = \epsilon Q$ with $\epsilon = \pm 1$; their orientation signs and origins compose. A monotone clock is necessarily unbounded on the lifted phase line. This is compatible with orbital stability on the compact phase circle and with modulated stability, which are separate notions. An emitter–receiver frequency ratio is phase-origin independent only when both endpoints use the same primitive orientation and origin covariance, both clock slopes are nonzero, the signal phase shifts affinely, and the ratio has total global charge zero.

This theorem is conditional in precisely one place: a nonzero observable requires a receiver cohomology class with nonzero descended pairing or period on a regular clock level. The counterflow charged-time fixture supplies the action–angle algebra. The later repaired- q_{70} health assembly now forms the physical quotients for every m, k at $j = 0, \frac{1}{2}, 1$. Their pairings are nonradical, but every certified block is physically unstable: respectively a real exponential direction, a Hamiltonian–Hopf

quartet and complex-frequency sectors. These instabilities survive fixed charge, while fixed- Q_{rel} reduction removes the relative clock itself. For $j \geq \frac{3}{2}$ the physical quotient and pairing remain `NO_CERTIFIED_MAP`. Thus no common healthy physical clock–receiver carrier exists in the certified domain, the thirteen-field receiver interface is not activated, and no detector or redshift value is inferred. The typed consumer result is `COUNTERFLOW_CHARGED_TIME_PHYSICAL_INSTANTIATION_AFTER_REPAIRED_Q70_HEALTH_NOT_ACTIVATED_V1`. The underlying conditional event-map theorem remains `CHARGED_PHYSICAL_TIME_RELATIONAL_EVENT_MAP_THEOREM_V1`.

Finite clock resolution can be introduced without changing that claim boundary. For an even normalized profile $\kappa \in C_c^\infty(\mathbb{R})$, set $\kappa_\epsilon(x) = \epsilon^{-1}\kappa(x/\epsilon)$ and

$$\mathcal{O}_A^\epsilon(\tau) = \int_M \kappa_\epsilon(\psi - \tau) d\psi \wedge A. \quad (76)$$

This is a test-function superposition of the distributional event currents, so BV closure, representative independence and the charged Q , H and K brackets descend unchanged. It remains inside the causal support of A and the clock band $|\psi - \tau| \leq \epsilon r_\kappa$; a profile tail may not be used to claim support preservation. As $\epsilon \rightarrow 0^+$ the family converges in the declared de Rham-current test class. If the receiver period P is twice differentiable, then

$$|S_\epsilon P(\tau) - P(\tau)| \leq \frac{\epsilon^2 \mu_2(\kappa)}{2} \sup_{|t-\tau| \leq \epsilon r_\kappa} |P''(t)|. \quad (77)$$

Profile second moments and support radii add under convolution. Two sampled phase transports with individual errors E_1, E_2 and outer Lipschitz constant L_2 obey the exact composition bound $E_2 + L_2 E_1$. Periodizing the profile gives compact-phase sampling, not a globally monotone real clock; nor does smoothing restore the clock removed by fixed-charge reduction. The counterflow receiver therefore remains `NO_CERTIFIED_MAP`. These statements are recorded in `CHARGED_TIME_FINITE_RESOLUTION_SAMPLING_THEOREM_V1`.

Conditional frequency comparisons also have an exact path algebra. For a declared retarded edge $i \rightarrow j$, let $C_{ij} = \nu_i^{(ij)} / \nu_j^{(ij)}$, assuming both sampled records exist and the denominator is bounded away from zero. Under independent primitive clock reversals it transforms by $\eta a_i / \eta_j$. Thus internal orientation factors cancel in a chain, but an edge is a scalar only after its endpoint orientation lines are identified. Reciprocity $C_{ji} = C_{ij}^{-1}$ uses the same transported record and a certified reverse path; the algebraic inverse does not turn an advanced path into a physical signal.

For three clocks the matched law is $C_{12}C_{23} = C_{13}$. Before matching, with endpoint records (a_1, a_2) , (b_2, b_3) and (c_1, c_3) , the exact defect is

$$\Delta_{123} = \frac{C_{12}C_{23}}{C_{13}} = \frac{a_1 b_2 c_3}{a_2 b_3 c_1}. \quad (78)$$

It reduces to the internal mismatch b_2/a_2 after endpoint identification. Without charge-fibre crosswalks the product is undefined, rather than a nontrivial observable. Around a labelled loop the comparison product equals the product of the vertex fibre/orientation transitions. The raw D , every $R_i = X_{Q_i}$ and every $K_i = D_i - H'_i(Q_i)R_i$ act separately.

Linear charge and support identities survive finite-resolution smearing, but smoothing does not commute with a ratio or product. Comparisons must be formed after sampling, with exact internal-record matching and nonzero denominator margins. These laws remain conditional until nonradical receiver periods exist, and no comparison is called a physical redshift. The exact ledger is `CHARGED_TIME_EMITTER_RECEIVER_COMPOSITION_THEOREM_V1`.

The conditional constructions admit a single fail-closed physical-receiver interface. A populated receiver consists of a compact local BV class $[A]$, a descended pairing with a nonradical witness and nonzero period, a support-preserving retarded chain map, a regular monotone clock interval, a positive sampled denominator margin, and separately typed D , R and K actions. A charge-fibre crosswalk must explicitly give its source and target fibres, charge and receiver-class maps, primitive orientation and origin change, endpoint-record identification, retarded path, and action intertwining. These conditions are necessary and sufficient within the declared current category for representative-independent evaluation, finite-resolution convergence, scoped reciprocity, matched composition and loop holonomy.

Failures have different meanings. An exact class or zero period gives a zero response; a radical class or vanishing denominator makes the response undefined; a missing fibre map is `NO_CERTIFIED_MAP`; and an advanced-only inverse loses physical causal interpretation. Fixing and reducing Q_{rel} remains an obstruction because it removes the clock. The machine-readable census covers all five observer atlas rows present at the import snapshot: three are conditional interfaces, the unrestricted counterflow row lacks its physical receiver, and the fixed-charge row has no clock. Thus no current row is promoted to an action-derived nonzero receiver. The population contract is `CHARGED_TIME_PHYSICAL_RECEIVER_CROSSWALK_INTERFACE_V1`, certified by `CHARGED_TIME_RECEIVER_ADMISSIBILITY_CROSSWALK_V1`.

The legacy Berger apparatus certificates have been replayed against this contract. Three exact matrices—the dynamical-emitter, localized-current and homogeneous-Maxwell source-to-probe transfers—retain rank two in their original pre-quotient carriers. This rank does not descend by itself to a physical detector algebra. None of the seven audited rows supplies, on one action-derived carrier, a detector BV class and cocycle, its receiver quotient, a descended nonradical period, a sampled denominator margin, and separately typed D , R and K actions. The exported advanced Green map is used only as the adjoint selecting emitter Cauchy data; it is not a signal. The quartic completion locus remains empty, so it supplies no nonlinear redshift carrier. Accordingly the exact census `BERGER_LEGACY_RECEIVER_ADMISSIBILITY_REPLAY_V1` promotes no physical receiver or redshift while preserving each linear response theorem in its original scope.

There is therefore no operational frequency ratio on the legacy census. The three maximal nonempty linear chains retain rank-two retarded response only as source-to-probe maps: the massive-two-form emitter chain, the localized external-Maxwell-current chain and the homogeneous Maxwell chain. None has an action-derived receiver BV cocycle and unary row from which a receiver quotient, nonradical period and positive sampled denominator could be constructed. In the homogeneous control sector both coordinate frequencies are $2\sqrt{10}/3$, giving the exact coordinate ratio one; this is not a relational redshift. The nonactivation theorem `BERGER_LEGACY_RECEIVER_OPERATIONAL_FREQUENCY_RATIO_NONACTIVATION_V1` made no operational ratio claim and issued one minimal producer request for an action-derived local receiver cocycle. That request has since been answered, without activating the ratio. A standalone $D0$ receiver action has the exact local descent $s(m^+) + d(\lambda\rho_0 d^3 R) = 0$, but ambient integration is obstructed. After a fresh rederivation in the repaired q_{70} compact grading, the local BV differential has degree -1 , the ambient action-Hessian chain has degree $+1$, and no degree-zero inclusion can satisfy $q_{70}i = is$ on the nonzero memory-shift descent. Hence no receiver quotient, nonradical period or operational frequency-ratio domain is produced. This is the Phase 1 exact-obstruction endpoint; it does not authorize a suspension bridge or another receiver architecture. The machine-readable crosswalk is `PHASE1_RELATIONAL_OBSERVABLE_DISPOSITION_SYNTHESIS_V1`.

References

- [1] R. Bach, *Zur Weylschen Relativitätstheorie und der Weylschen Erweiterung des Krümmungstensorbegriffs*, Math. Z. **9** (1921), 110–135, doi:10.1007/bf01378338.
- [2] I. A. Batalin and G. A. Vilkovisky, *Gauge algebra and quantization*, Phys. Lett. B **102** (1981), 27–31, doi:10.1016/0370-2693(81)90205-7.
- [3] M. Henneaux and C. Teitelboim, *Quantization of Gauge Systems*, Princeton University Press, 1992, doi:10.1515/9780691213866.
- [4] C. Rovelli, *Partial observables*, Phys. Rev. D **65** (2002), 124013, doi:10.1103/physrevd.65.124013.
- [5] B. Dittrich, *Partial and complete observables for Hamiltonian constrained systems*, Gen. Relativ. Gravit. **39** (2007), 1891–1927, doi:10.1007/s10714-007-0495-2.
- [6] C. Goeller, P. A. Höhn, and J. Kirklin, *Diffeomorphism-invariant observables and dynamical frames in gravity: reconciling bulk locality with general covariance*, arXiv:2206.01193.
- [7] C. Crnkowic and E. Witten, *Covariant description of canonical formalism in geometrical theories*, in *Three Hundred Years of Gravitation*, Cambridge University Press, 1987.
- [8] I. Khavkine, *Covariant phase space, constraints, gauge and the Peierls formula*, Int. J. Mod. Phys. A **29** (2014), 1430009, doi:10.1142/s0217751x14300099.
- [9] A. E. Fischer and J. E. Marsden, *Linearization stability of the Einstein equations*, Bull. Amer. Math. Soc. **79** (1973), 997–1003, doi:10.1090/s0002-9904-1973-13299-9.
- [10] J. M. Arms, J. E. Marsden, and V. Moncrief, *The structure of the space of solutions of Einstein’s equations II: several Killing fields and the Einstein–Yang–Mills equations*, Ann. Phys. **144** (1982), 81–106, doi:10.1016/0003-4916(82)90105-1.
- [11] A. S. Cattaneo, P. Mnev, and N. Reshetikhin, *Classical and quantum Lagrangian field theories with boundary*, PoS CORFU2011 (2012), 044, doi:10.22323/1.155.0044.
- [12] H. Kajiwara, *Noncommutative homotopy algebras associated with open strings*, Rev. Math. Phys. **19** (2007), 1–99, doi:10.1142/s0129055x07002912.
- [13] A. S. Arvanitakis, O. Hohm, C. Hull, and V. Lekeu, *Homotopy transfer and effective field theory I: tree level*, arXiv:2007.07942.
- [14] D. Fiorenza and M. Manetti, *L_∞ algebras, Cartan homotopies and period maps*, arXiv:math/0605297.
- [15] A. Kraft and J. Schnitzer, *An introduction to L_∞ -algebras and their homotopy theory*, arXiv:2207.01861.
- [16] C. G. Callan, S. Coleman, and R. Jackiw, *A new improved energy–momentum tensor*, Ann. Phys. **59** (1970), 42–73, doi:10.1016/0003-4916(70)90394-5.
- [17] M. Benini, G. Musante, and A. Schenkel, *Green hyperbolic complexes on Lorentzian manifolds*, Commun. Math. Phys. (2023), doi:10.1007/s00220-023-04807-5.